Code-Layout, Readability and Reusability

Date	06 July 2026
Team ID	TMID-SL-2026-01
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

This document evaluates the quality of the codebase in terms of structure, readability, and reusability.

S.No	Code Quality Parameter	Description	Followed	Remarks
1	**Consistent Indentation**	Uniform spacing/tabs used throughout the code	Yes	PEP8-style 4-space indentation across app.py and train_model.py
2	**Proper File Structure**	Files and folders are logically organized	Yes	Dataset/, Training/, Flask/ separated by concern
3	**Meaningful Variable Names**	Variables reflect their purpose clearly	Yes	e.g. x_bal, y_bal, X_train, scaler
4	**Function / Method Names**	Functions are descriptively named	Yes	handle_categorical_and_missing(), balance_data(), evaluate()
5	**Code Comments**	Inline and block comments explain logic	Yes	Docstrings on train_model.py and app.py, inline comments on key steps
6	**Modular Design**	Code is split into reusable functions/modules	Yes	Preprocessing, balancing, scaling, evaluation each isolated as functions
7	**No Redundant Code**	Duplicate or unused code is removed	Yes	Shared evaluate() function used across all four models
8	**Error Handling**	Exceptions and errors are handled gracefully	Partial	Flask route uses form.get() with defaults; broader try/except could be added for production

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability
1	evaluate()	Python / scikit-learn	Called for all four models (DT, RF, KNN, XGBoost)	High
2	handle_categorical_and_missing()	Python / pandas	Training pipeline; mirrored in app.py's encoding maps	High
3	style.css design tokens	CSS	Shared across home.html, predict.html, submit.html	High
4	FEATURE_ORDER list	Python	Keeps app.py's inference input aligned with training feature order	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of data, training, and app layers
Readability	4	Descriptive names and comments throughout
Reusability	4	Shared evaluation/preprocessing functions

Aspect	Rating (1-5)	Comments
Documentation / Comments	4	README + docstrings cover setup and pipeline
Overall Score	4.25	Solid, maintainable structure for a project of this scope